

Design and Finite-Element Comparison of Orthopedic Fracture Fixation Plates Under Jump-Landing Loads

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Abstract

Fracture fixation plates must withstand the loads generated by a patient's everyday activity. This study combined ground reaction force (GRF) data from a deep-squat jump landing (peak 3073.27 N; 1536.635 N applied per leg) with static finite-element simulations in SolidWorks to compare three fixation plate designs within an $8 \times 2 \times 1$ cm envelope. The plates were made of Ti-6Al-4V, alloy steel and 1060 aluminium alloy, with two 5 mm or three 7 mm screw holes. Design 1 (Ti-6Al-4V, two 5 mm holes) had the lowest maximum von Mises stress (0.7233 MPa), displacement (1.692×10^{-3} mm) and strain (4.394×10^{-6}). Design 3 (1060 alloy, three 7 mm holes) reached 210.1 MPa, about 7.6 times its yield strength, indicating it would yield under this load. Because the plates differed in geometry as well as material, and the loading was static and simplified, the results are a preliminary comparison rather than a validated ranking.

Keywords: *fracture fixation, orthopedic plate, finite-element analysis, SolidWorks Simulation, ground reaction force, Ti-6Al-4V*

1. Introduction

Fracture fixation plates are widely used orthopedic devices that stabilise broken bones and maintain alignment during healing. They must withstand the mechanical loads a patient produces during physical activity, so their design depends on strength and stiffness and on an understanding of how forces act on the tibia, including the ground reaction forces generated when landing from a jump.

CAD modelling and simulation let engineers explore many plate designs, sizes and materials before manufacturing anything. In this project, three plates were designed in SolidWorks from Ti-6Al-4V, alloy steel and 1060 aluminium alloy, and each was tested in a static simulation under the same landing load. The objectives were to:

- derive a realistic peak loading condition from measured jump-landing GRF data;
- model three fixation plates of different materials and screw-hole configurations; and
- compare stress, displacement and strain across the designs to identify the best-performing plate.

2. Methods

2.1 Jump Data Collection

The jump data came from the procedure used in Lab 3. A participant performed several vertical jumps on a force plate while being recorded from the side, keeping their hands on their hips to limit arm motion. A reflective marker on the greater trochanter approximated the centre of mass (COM). The video was calibrated with a known reference object, and GRF was recorded at 500 Hz after zeroing the plate. Each jump started and ended on the plate so that takeoff, flight and landing were all captured (Figure 1).



Figure 1. *Experimental setup for the jump measurements: force plate with a side-view camera on a tripod.*

2.2 Peak Landing Force

The vertical GRF was filtered in MATLAB to remove plate noise. COM acceleration was calculated by subtracting body weight from the vertical force and dividing by body mass, and integrating this once and twice gave velocity and displacement, which identified the takeoff, flight and landing phases. During landing the COM decelerates quickly and the GRF rises sharply above body weight. The highest value in this region was defined as the peak landing force.

Because the participant, jump type and setup were unchanged, the peak landing force from Lab 3 was reused rather than repeating the experiment. For simulation, the force was assumed to act along the tibial axis at the mid-shaft, and since the load is shared between both legs, half of the peak value was applied to the plate. This gave one consistent loading condition for all designs, although joint angles and soft tissue would distribute the load differently in reality.

2.3 Plate Design

The plates were modelled in SolidWorks within the handout's dimensional limit of $8 \times 2 \times 1$ cm, with screw holes of 5 mm or 7 mm diameter. Screw files were imported and arranged in a configuration unique to each group member, and each plate was assigned a material from the recommended list (Table 1).

Table 1. *Design parameters.*

Design	Material	Screw holes	Hole diameter (mm)	Yield strength (MPa)
1	Ti-6Al-4V, solution treated and aged	2	5	827.4
2	Alloy steel	2	5	620.4
3	1060 aluminium alloy	3	7	27.57

2.4 Finite-Element Simulation

Each plate was assembled with its screws and analysed with the SolidWorks static study tool. The peak landing force (1536.635 N) was applied to the superior screw, and the inferior screw was fixed to represent the constraint from bone. The model was then meshed and solved to give maps of von Mises stress, resultant displacement and equivalent strain, so that all designs could be compared under the same landing load (Figure 2).

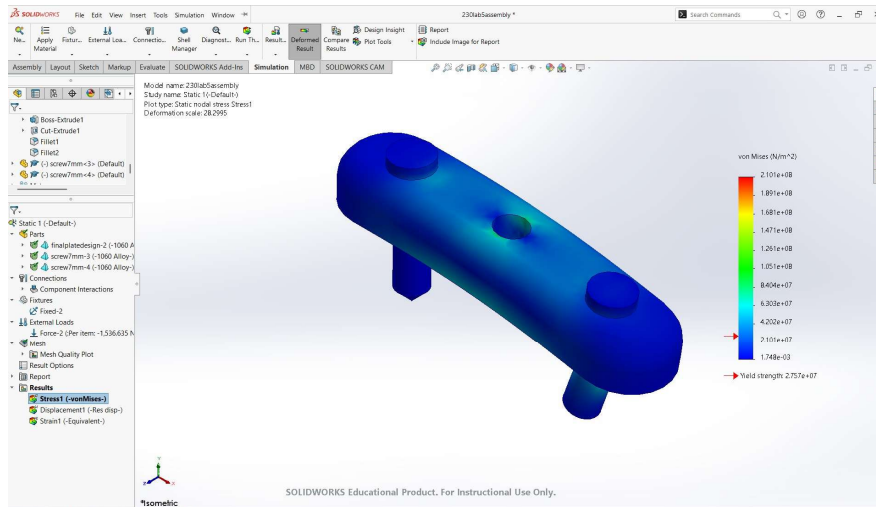


Figure 2. Static study set-up and stress result in SolidWorks Simulation, shown for Design 3. The red arrow on the colour scale marks the material's yield strength.

3. Results

3.1 Loading Condition

Figure 3 shows the vertical GRF for the deep-squat jump. Force drops to zero during flight and peaks on landing at 3073.27 N, about 3.8 times the body weight of the 82 kg participant. Half of this value, 1536.635 N, acts through one leg and is the load used in the simulations (Figure 4).

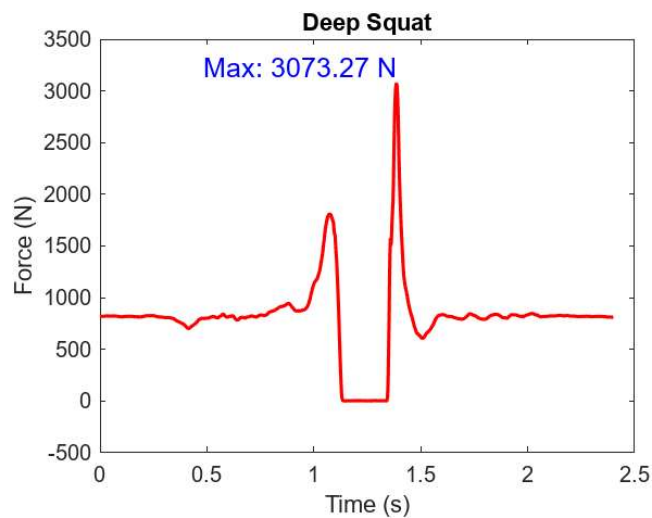


Figure 3. Vertical ground reaction force over time for the deep-squat jump, with the peak landing force of 3073.27 N.

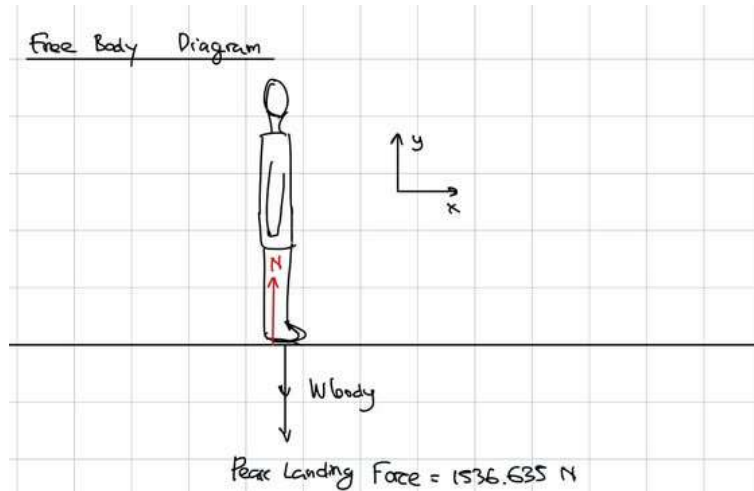


Figure 4. Free-body diagram of the landing, showing body weight, the normal reaction and the peak landing force applied per leg (1536.635 N).

3.2 Plate Designs

Figure 5 shows the SolidWorks drawing of Design 1 and Figure 6 shows all three designs.

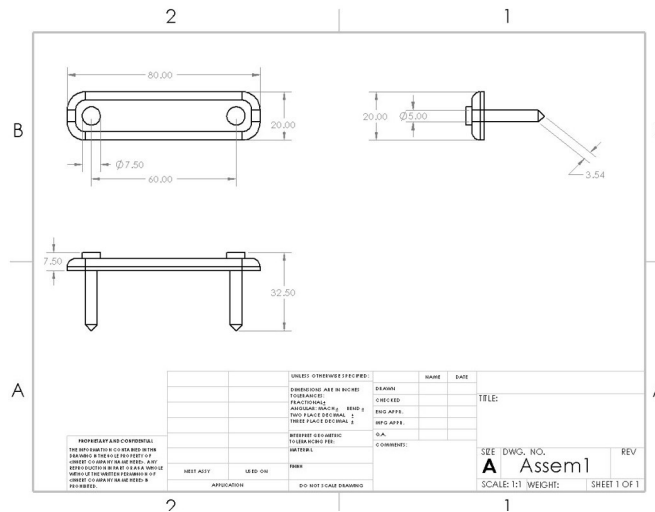
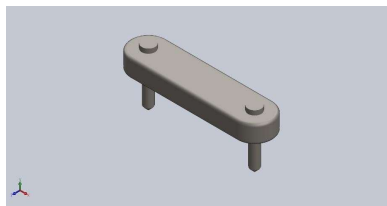


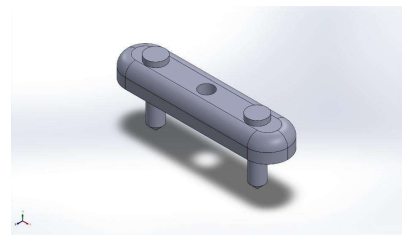
Figure 5. SolidWorks drawing of Design 1 (top, side and front views; dimensions in mm).



Design 1



Design 2



Design 3

Figure 6. The three plate designs: Design 1 (Ti-6Al-4V, two 5 mm holes), Design 2 (alloy steel, two 5 mm holes) and Design 3 (1060 alloy, three 7 mm holes).

3.3 Simulation Results

Table 2 summarises the results. Design 1 had the lowest stress, displacement and strain, and Figure 7 shows its stress distribution. Design 3 had the highest displacement and strain, and its maximum stress exceeded the yield strength of 1060 alloy (Figure 2). The factor of safety in Table 2 is the yield strength divided by the maximum von Mises stress; a value below 1 means the material yields.

Table 2. Simulation results under the 1536.635 N landing load.

Design	Max von Mises stress (MPa)	Max displacement (mm)	Equivalent strain	Factor of safety
1	0.7233	1.692×10^{-3}	4.394×10^{-6}	≈ 1140
2	177.0	5.545×10^{-2}	4.719×10^{-4}	≈ 3.5
3	210.1	3.242×10^{-1}	2.093×10^{-3}	≈ 0.13

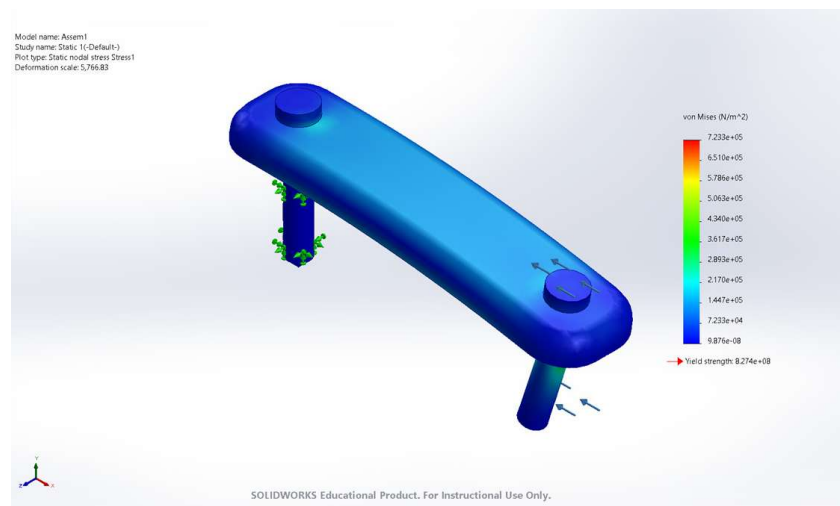


Figure 7. Von Mises stress in the best-performing design, Design 1 (Ti-6Al-4V). The maximum stress is 7.233×10^5 N/m² against a yield strength of 8.274×10^8 N/m².

4. Discussion

4.1 Design Comparison

Design 1 performed best on every metric. Its maximum stress was roughly 245 times lower than Design 2 and 290 times lower than Design 3, and it also had the smallest displacement and strain. Design 3 was the weakest: its peak stress of 210.1 MPa is about 7.6 times the 27.57 MPa yield strength of 1060 alloy, so this plate would deform permanently under a single landing. Its stress and strain values should therefore be read as a sign of failure rather than as quantitatively reliable, since a linear static study does not model yielding. Design 2 stayed within its yield strength with a factor of safety of about 3.5.

4.2 Role of Material

Ti-6Al-4V has the highest yield strength of the three materials (827.4 MPa, compared with 620.4 MPa for alloy steel and 27.57 MPa for 1060 alloy) and a high strength-to-weight ratio. It is also widely used in orthopedic implants for its corrosion resistance and biocompatibility, and its lower stiffness than steel reduces stress shielding of the healing bone. 1060 alloy is commercially pure aluminium and has by far the lowest yield strength, which explains its large deformation.

The material alone does not explain the ordering, however. Alloy steel is stiffer than Ti-6Al-4V (a Young's modulus of roughly 210 GPa versus about 114 GPa), so with identical geometry and constraints it would deflect less, not more. Stress in a linear static analysis is also governed mainly by geometry, load and constraints rather than by material. The very large gap between Designs 1 and 2, which share the same hole count and diameter, therefore points to differences in plate geometry or boundary conditions between the models. Material and geometry effects cannot be separated with this data.

4.3 Role of Screw Holes

Holes concentrate stress near their edges and remove load-bearing material, and larger or more numerous holes reduce the plate's stiffness and strength. Design 3 has three 7 mm holes, the most material removed of any design, which is consistent with its high displacement and strain. It also uses the weakest material, so the two effects are confounded.

4.4 Limitations

- The loading was a single static peak force applied to one screw. Real loading is dynamic and repeated, is shared between the plate, bone and soft tissue, and acts in a direction that depends on posture after landing.
- The bone was represented only by a fixed inferior screw, with no fracture gap or bone–plate contact modelled.
- The load came from one participant (a 19-year-old, 82 kg male) landing from a deep squat, which is not a typical daily activity and does not represent the average patient.
- Each plate had its own geometry as well as its own material, so the comparison does not isolate the effect of either.

4.5 Future Work

Applying the full time-varying GRF curve in a dynamic study, and testing different loading angles and everyday activities such as walking and stair descent, would give a more realistic picture of plate performance. Simulating each geometry in every material would separate material from geometry effects, and a mesh-convergence check and data from more participants would strengthen the results.

5. Conclusion

This project combined jump-landing force data with SolidWorks static simulations to compare three fixation plate designs. The Ti-6Al-4V plate with two 5 mm holes (Design 1) gave the lowest stress, displacement and strain under a 1536.635 N landing load, while the 1060 alloy plate with three 7 mm holes (Design 3) exceeded its yield strength. The results show that material choice and screw-hole configuration both influence structural performance, but because the designs differed in geometry and the loading was static and simplified, the findings are best treated as an early design screen. They show how simulation can flag plate configurations unlikely to tolerate physiological loads before any testing.

Author Contributions

P. Khanvilkar: results, discussion and SolidWorks modelling. F. A. Wahyudi: methodology, conclusion and SolidWorks modelling. C. J. Setiawan: background, free-body diagram and SolidWorks modelling.